COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 5 — Spiral Model

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Section 12: Classroom Activity — Build a Risk Register

Context: Risk analysis and mitigation planning for a critical Hospital Patient Appointment & Emergency Notification System using Spiral Model principles.

Risk ID	Risk Description	Likelihood	Impact	Priority	Mitigation / Engineering Action
RSK-01	Breach of Confidential Patient Medical History (HIPAA / Data Protection Violation).	Medium	High	High	Enforce AES-256 field-level encryption on diagnostic data; mandate multi-factor authentication (MFA) for staff.
RSK-02	Double-Booking of Specialist Doctors due to concurrent booking collisions.	High	High	Critical	Implement strict database table row locks and Redis-based temporary appointment holding locks during checkout.
RSK-03	Emergency SMS/Voice notification failure during server network outage.	Low	High	High	Deploy redundant multi-carrier SMS providers with automatic failover fallback if delivery receipt latency exceeds 5 seconds.

Priority Justification: RSK-02 is categorized as Critical priority because duplicate surgical appointments disrupt clinical operations, cause patient distress, and lead to direct institutional liability.

Section 15: Homework — Risk Register for Online Banking Application

Comprehensive four-point risk register for a mission-critical Retail Online Banking Web Portal.

DAKSH

Risk Description	Likelihood	Impact	Engineering Mitigation / Spike Investigation	Measurable Evidence of Risk Reduction
Man-in-the-Middle (MitM) & Session Hijacking attacks on public networks.	Medium	High	Mandate Strict-Transport-Security (HSTS), certificate pinning in mobile apps, and 5-minute session inactivity timeouts.	Zero unencrypted HTTP requests accepted; penetration testing confirms cookie hijacking immunity.
Ledger inconsistency caused by distributed transaction failure between payment switch and core ledger.	Low	High	Implement Two-Phase Commit (2PC) or SAGA pattern with idempotent transactional message outbox queues.	10,000 simulated network drop transactions yield zero ledger imbalance discrepancies.
Mass login lockouts caused by credential-stuffing distributed brute-force attacks.	High	Medium	Deploy Cloudflare bot protection, adaptive rate limiting, and CAPTCHA challenge after 3 failed attempts per IP.	Brute-force script requests throttled at network edge with 0% impact on legitimate user logins.
Database hardware failure causing unrecoverable customer transaction history loss.	Low	High	Configure synchronous multi-AZ database clustering with continuous write-ahead log (WAL) streaming to cold storage.	Automated disaster recovery drill demonstrates RPO < 5 seconds and RTO < 60 seconds.